24ADI003-MACHINE LEARNING
UNIT 2: **REGRESSION MODELS**

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MULTIPLE LINEAR REGRESSION

- In **Multiple Linear Regression**, the word **linear** refers to:
- **Linearity with respect to the PARAMETERS (weights), not the variables.**

General Model

- $y = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n$

Here:

- $w_0, w_1, w_2,$ ...are **parameters (weights)**
- $x_1, x_2,$ ...are **input features**

The model is *linear* because:

- Every weight appears **only to power 1**
- Weights are **added**, not multiplied with each other

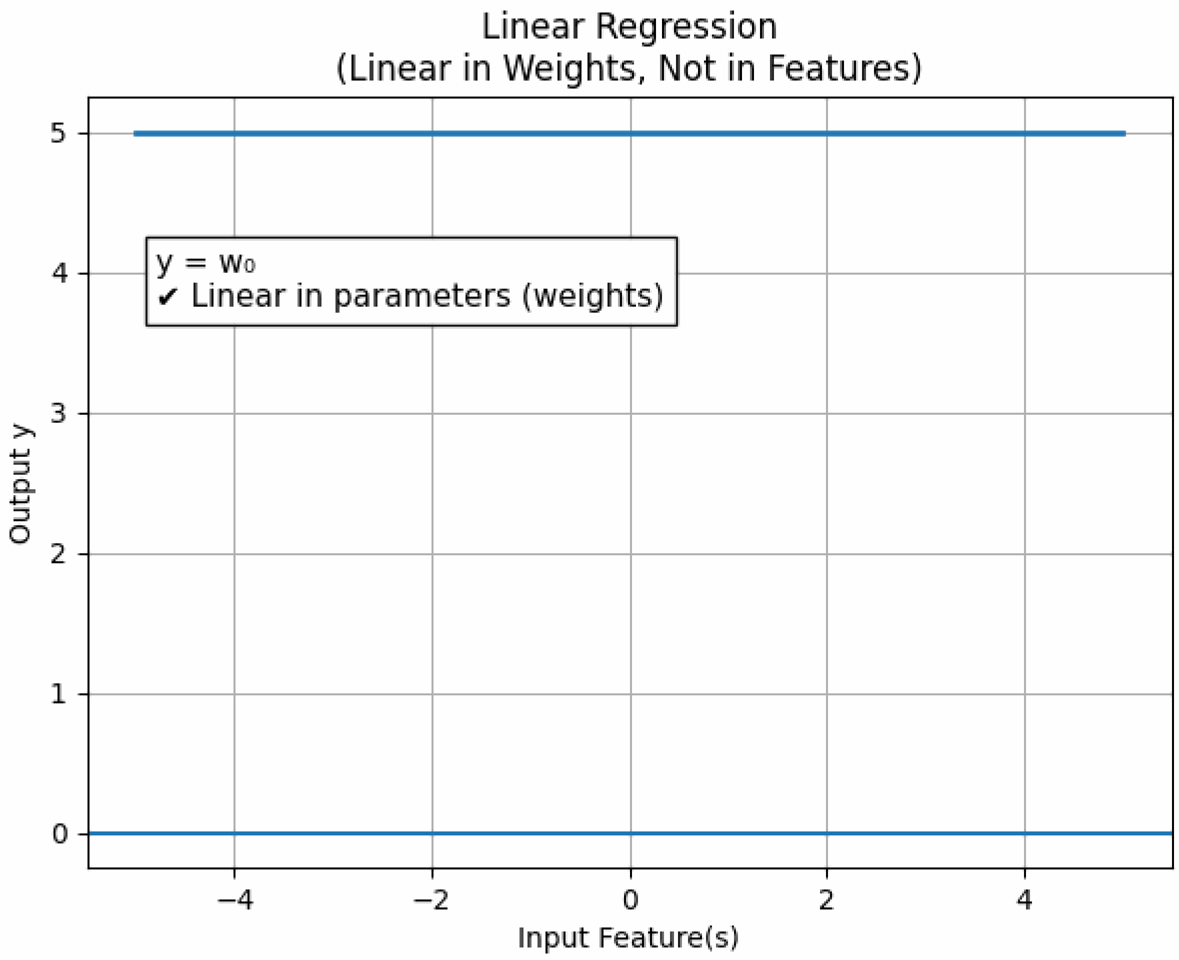
MULTIPLE LINEAR REGRESSION

THESE ARE Linear Regression

These are **linear models**:

- $y = 5 + 2x_1 + 3x_2$
- $y = w_0 + w_1x + w_2x^2$

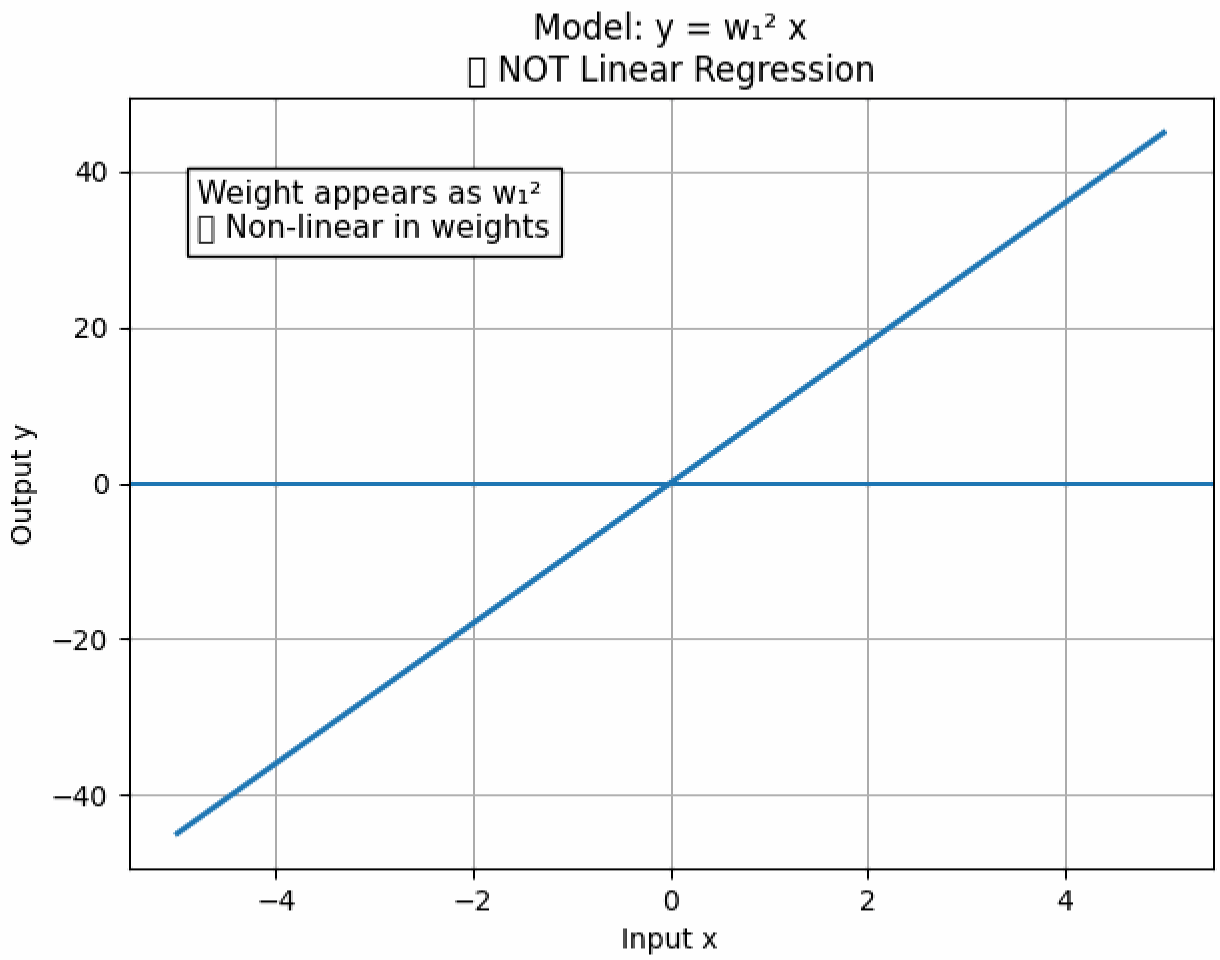
Even though x^2 exists, the **weights are still linear**.



MULTIPLE LINEAR REGRESSION

- These are **NOT linear models**:

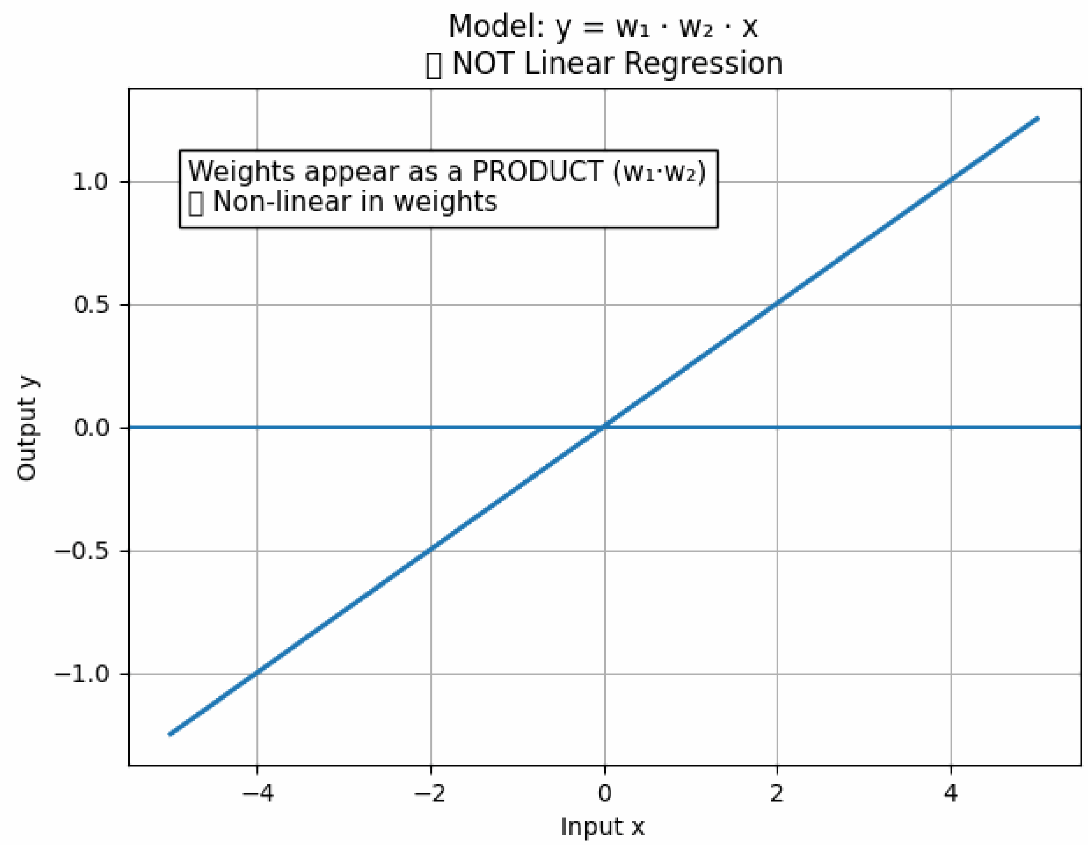
$$y = w_1^2 x$$



MULTIPLE LINEAR REGRESSION

- These are **NOT linear models**:

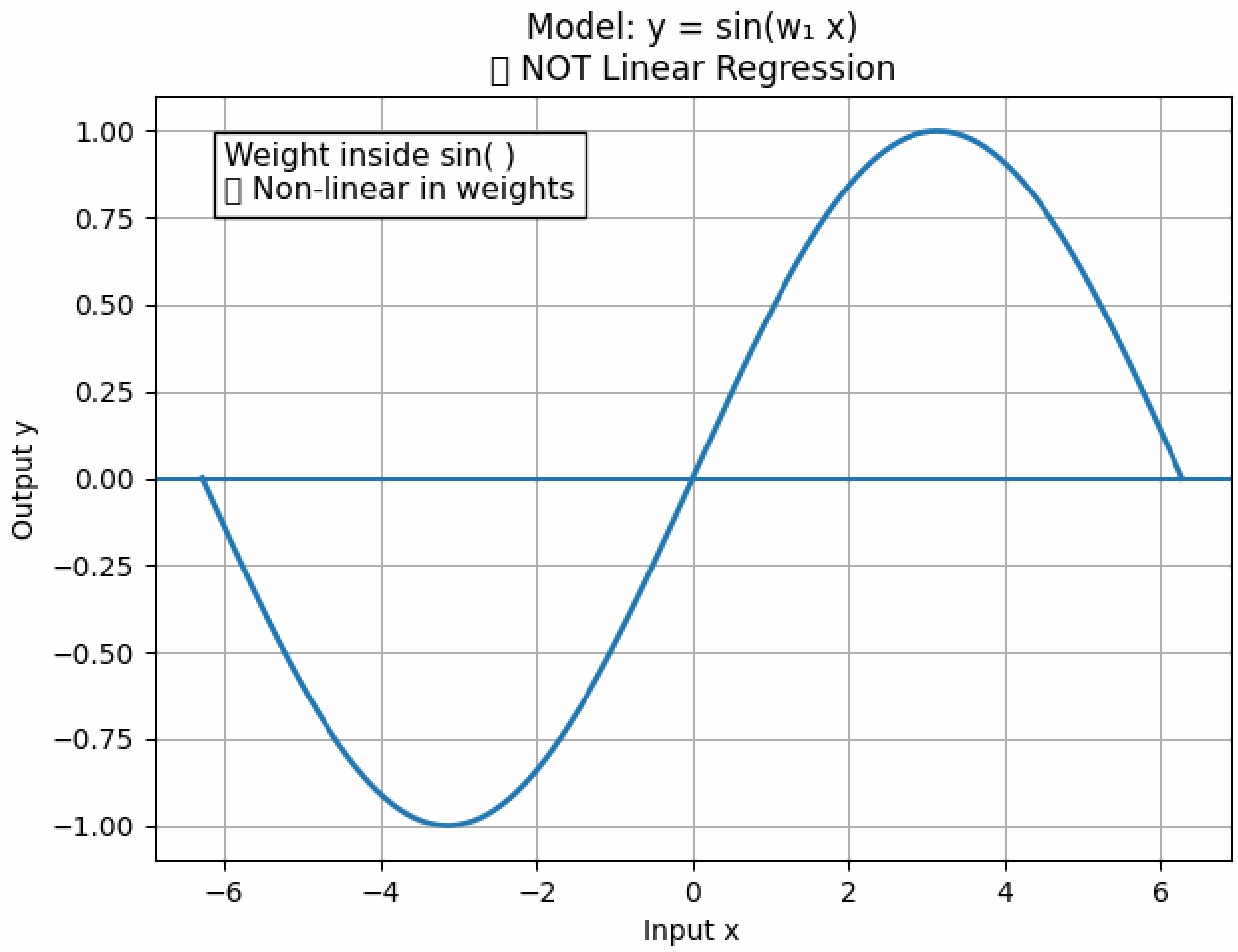
$$y = w_1 w_2 x$$



MULTIPLE LINEAR REGRESSION

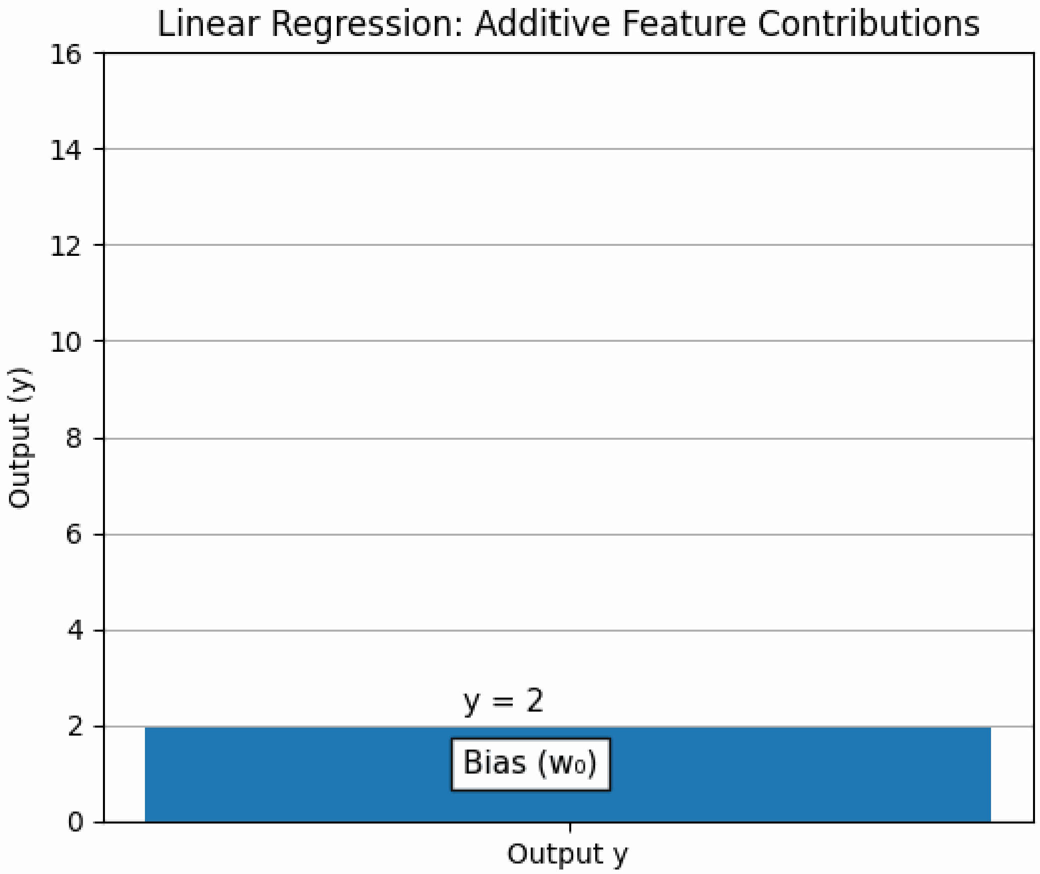
- These are **NOT linear models**:

$$y = \sin(w_1 x)$$



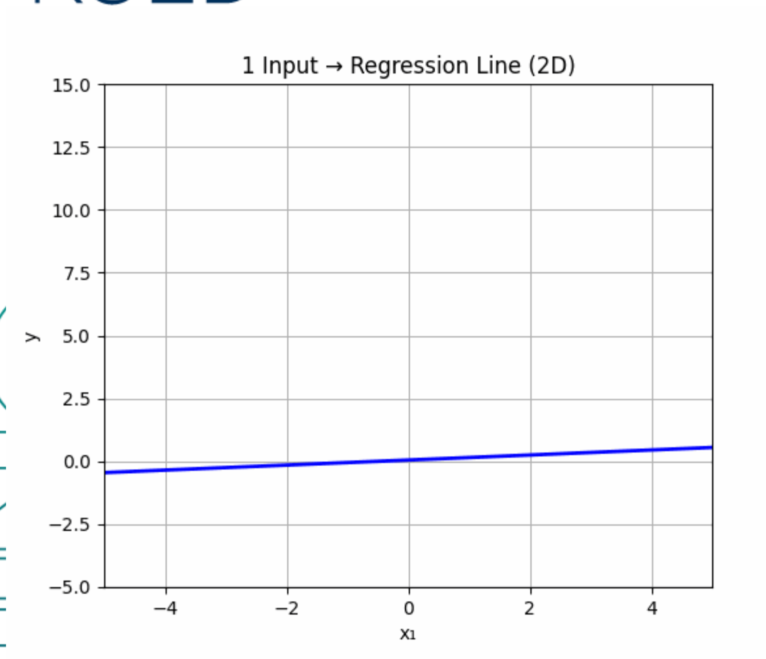
MULTIPLE LINEAR REGRESSION

- Output (y)
- ▲
- │ Contribution from x1 = w1 × x1
- │ Contribution from x2 = w2 × x2
- │ Contribution from x3 = w3 × x3
- │
- └──── y = w0 + (w1x1 + w2x2 + w3x3)

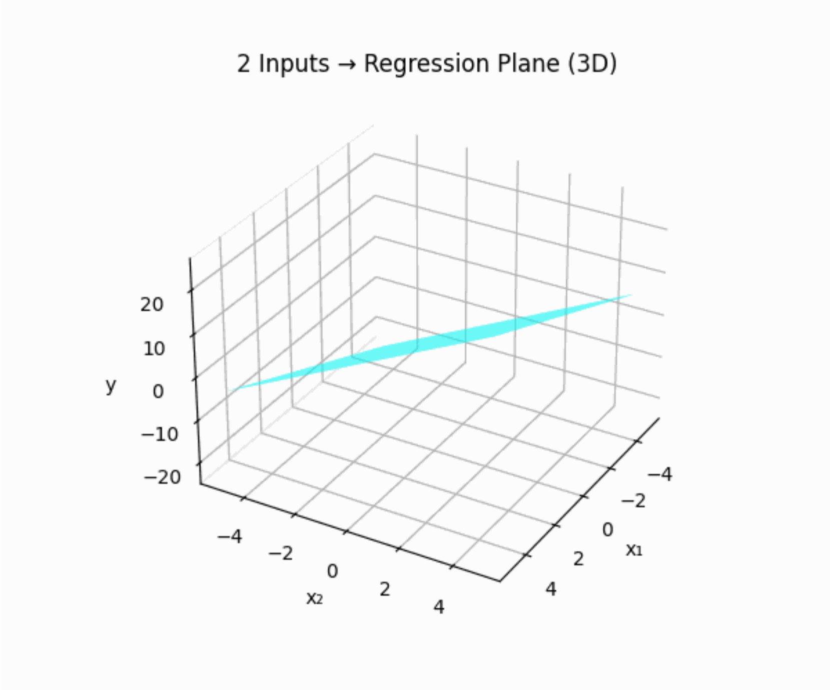


•Increase in **one feature** affects output **proportionally**
•Other features remain unchanged

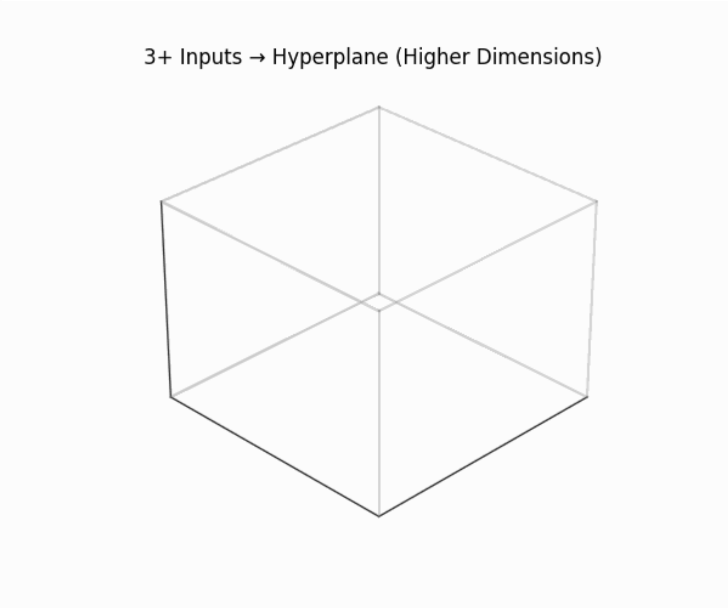
MULTIPLE LINEAR REGRESSION



$y = w_0 + w_1x_1$



$y = w_0 + w_1x_1 + w_2x_2$



$y = w_0 + w_1x_1 + w_2x_2 + w_3x_3$

MULTIPLE LINEAR REGRESSION

Concept of Multiple Linear Regression

- In many real-world situations, **one output depends on several factors**, not just one.

Examples:

- Student performance depends on **study hours, attendance, assignments**
- House price depends on **area, number of rooms, location**
- Salary depends on **experience, education, skills**

- **Multiple Linear Regression** models this relationship mathematically.

- Multiple Linear Regression is a supervised learning algorithm that models the relationship between one continuous dependent variable and two or more independent variables using a **linear equation in parameters**.

MULTIPLE LINEAR REGRESSION

Mathematical Model

$$\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n$$

Where:

- $\hat{y}$ →predicted output
- $x_1, x_2, \dots, x_n$ →input features
- $w_1, w_2, \dots, w_n$ →weights (importance of features)
- w_0 →bias (intercept)

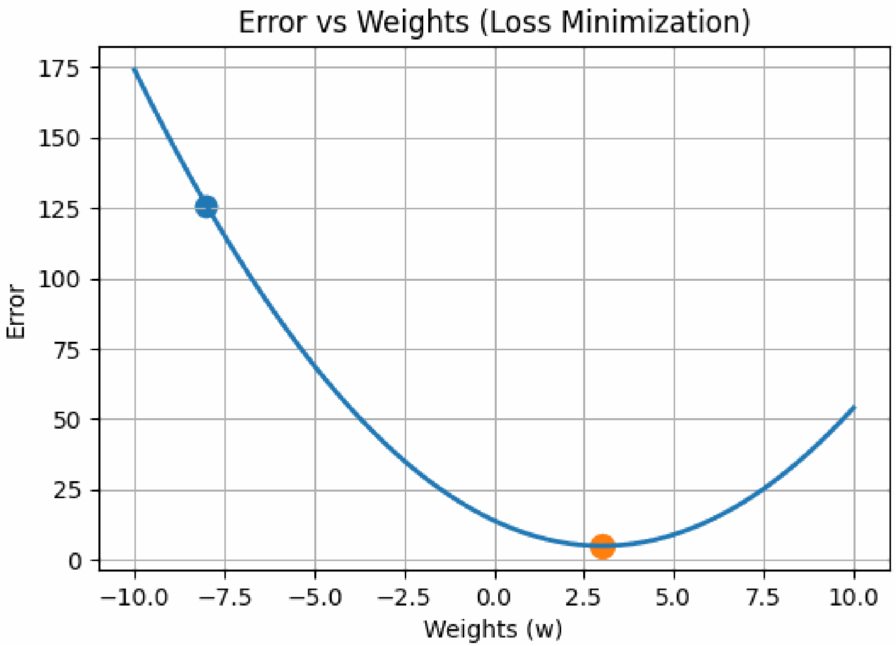
Final Output =

- (Base value)
- + (Effect of feature 1)
- + (Effect of feature 2)
- + …
- Each feature **contributes independently**, and all contributions are **added together**.

MULTIPLE LINEAR REGRESSION

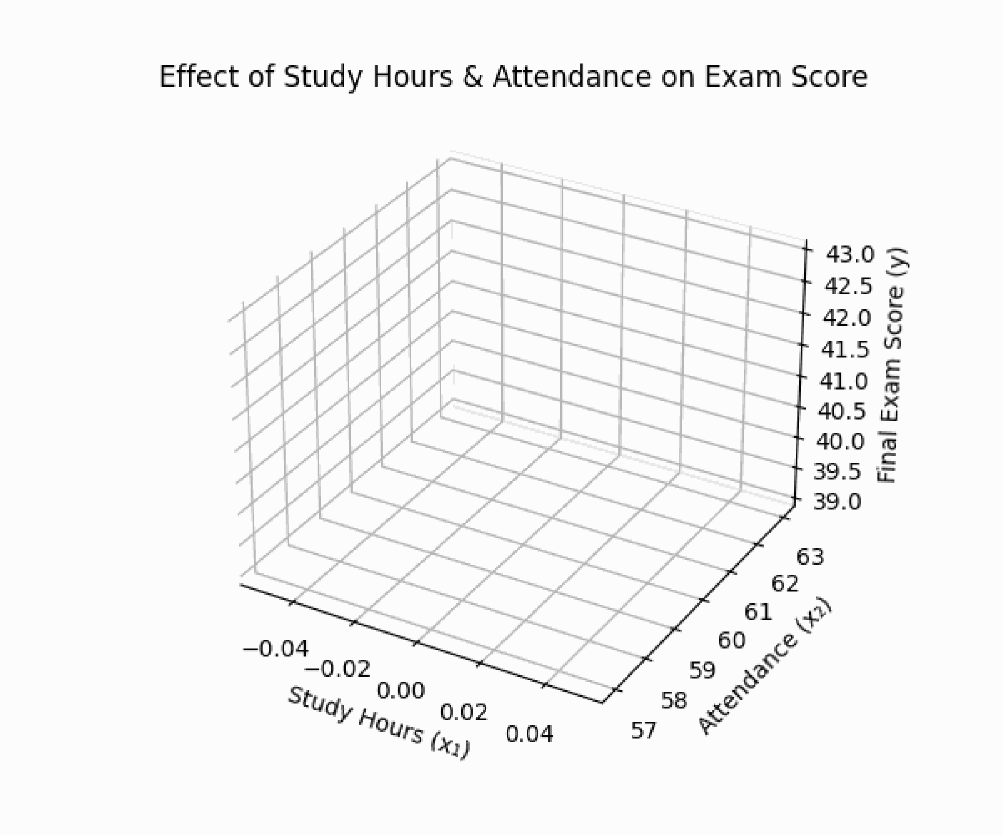
- The goal of Multiple Linear Regression is to **find the best values of weights** such that the prediction error is minimum.
- The most common error measure is **Mean Squared Error (MSE)**:

$$\mathrm{MSE} = \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2$$



MULTIPLE LINEAR REGRESSION

- **Multiple Linear Regression** is a supervised machine learning algorithm that models the relationship between **one dependent variable** and **two or more independent variables** by fitting a **straight line (or plane)** that minimizes prediction error.
- ***We try to predict one output by combining the effects of many inputs using suitable weights.***



We assume a simple linear model:

$$y = 5 + 4x_1 + 0.6x_2$$

- x_1 :Study Hours
- x_2 :Attendance (%)
- y: Final Exam Score

MULTIPLE LINEAR REGRESSION

```
x1 (Hours) ─┐
 ├───▶ Linear Combination ──▶ y (Score)
x2 (Attendance) ─┘  w1·x1 + w2·x2 + w0
```

- Each input feature contributes **independently and additively** to the output.
    - w_1 controls influence of x_1
    - w_2 controls influence of x_2
    - w_0 shifts the prediction up or down
- Think of multiple regression as **adding weighted effects of many causes**.

MULTIPLE LINEAR REGRESSION

- Multiple Linear Regression is an extension of **Simple Linear Regression** where **more than one independent variable** is used to predict a dependent variable.
- **Simple Regression:**

$$y = w_0 + w_1 x$$

- Multiple $y = w_0 + w_1 x$ **Regression:**

$$y = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n$$

Here:

- y → Dependent (target) variable
- $x_1, x_2, \ldots, x_n$ →Independent (input) variables
- w_0 →Bias (intercept)
- $w_1, w_2, \ldots, w_n$ →Weights (coefficients)

MULTIPLE LINEAR REGRESSION

1Matrix Representation
To simplify calculations, we use **matrix notation**:

$$\mathbf{y} = \mathbf{X}\mathbf{w}$$

Where:

$\mathbf{X}$ → Design matrix (with a column of 1s for bias)

$\mathbf{w}$ → Weight vector

$\mathbf{y}$ → Output vector

MULTIPLE LINEAR REGRESSION

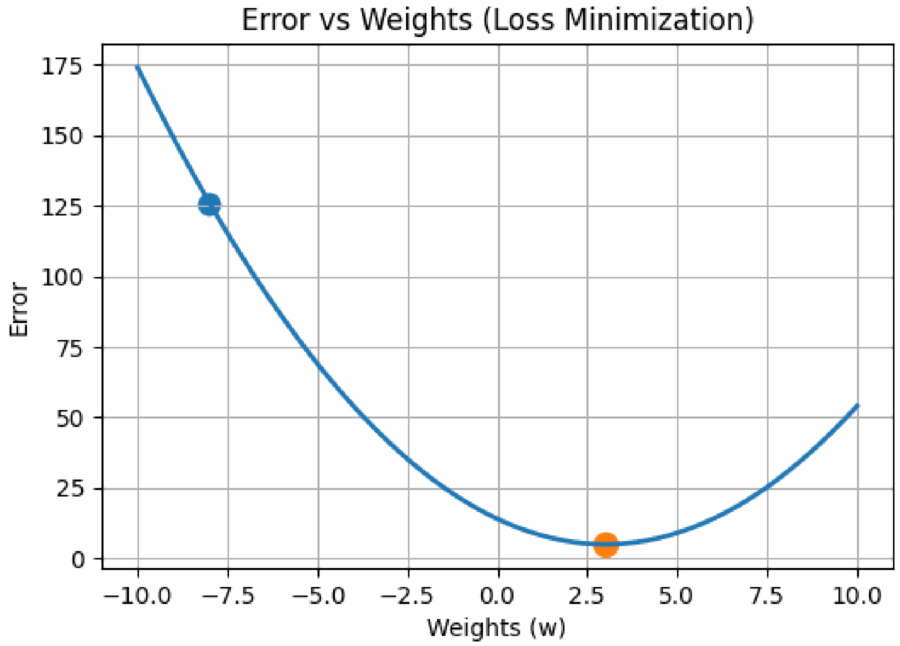
Objective of Multiple Regression:
The learning algorithm searches for the **lowest point** on the error surface.

We want to find weights **w** that minimize the **Mean Squared Error (MSE)**:

$$J(\mathbf{w}) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

This leads to the **Normal Equation**:

$$\mathbf{w} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$



MULTIPLE LINEAR REGRESSION

Problem Statement

A teacher wants to predict a student's **final exam score (y)** based on:

- x_1 =Study hours
- x_2 =Attendance percentage

Student	Study Hours (x_1)	Attendance % (x_2)	Score (y)
1	2	75	65
2	4	80	70
3	6	85	78

Step 1: Assume the Regression Model

$$\hat{y} = w_0 + w_1 x_1 + w_2 x_2$$

MULTIPLE LINEAR REGRESSION

Problem Statement

Step 2: Construct the Design Matrix

Add a column of 1s for the bias term.

$$X = \begin{bmatrix} 1 & 2 & 75 \\ 1 & 4 & 80 \\ 1 & 6 & 85 \end{bmatrix}, \quad Y = \begin{bmatrix} 65 \\ 70 \\ 78 \end{bmatrix}$$

[1 x1 x2] → Input with bias

Step 3: Normal Equation

To find optimal weights:

$$W = (X^T X)^{-1} X^T Y$$

MULTIPLE LINEAR REGRESSION

Problem Statement

Step 4: Compute $X^T X$

$$X^T X = \begin{bmatrix} 3 & 12 & 240 \\ 12 & 56 & 980 \\ 240 & 980 & 19250 \end{bmatrix}$$

Step 5: Compute $X^T Y$

$$X^T Y = \begin{bmatrix} 213 \\ 928 \\ 17165 \end{bmatrix}$$

Step 6: Compute Weights

After calculating $(X^T X)^{-1} X^T Y$:

$$W = \begin{bmatrix} 10 \\ 2.5 \\ 0.4 \end{bmatrix}$$

So:

- $w_0 = 10$
- $w_1 = 2.5$
- $w_2 = 0.4$

MULTIPLE LINEAR REGRESSION

Problem Statement

Step 7: Final Regression Equation

$$\boxed{\hat{y} = 10 + 2.5x_1 + 0.4x_2}$$

Step 8: Interpretation of the Model

- **Bias (10):** Base score even with zero inputs
- **Study hours coefficient (2.5):**
  → Each additional study hour increases score by **2.5 marks**
- **Attendance coefficient (0.4):**
  → Each 1% increase in attendance increases score by **0.4 marks**
- *All interpretations assume other variables are constant.*

POLYNOMIAL REGRESSION

Polynomial Regression is a regression technique that models the relationship between input and output as a **polynomial curve** instead of a straight line.

- Polynomial Regression models the dependent variable as a polynomial function of the independent variable.

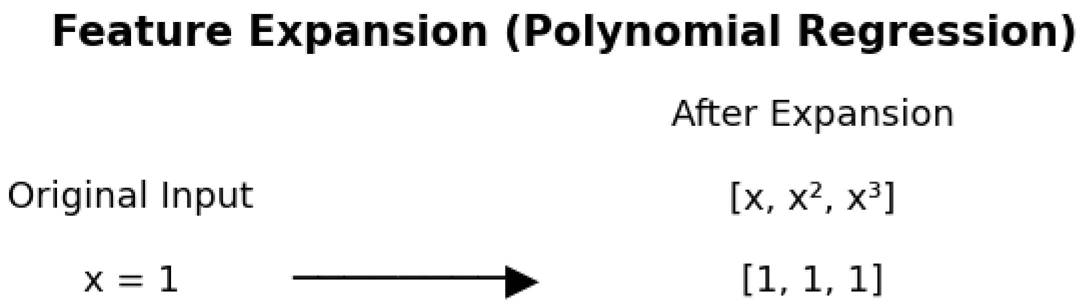
$$\hat{y} = w_0 + w_1 x + w_2 x^2 + \cdots + w_d x^d$$

- Polynomial regression does **NOT** invent a new algorithm.

Instead, it:

- **Transforms the input features.**
- Applies **linear regression** on the transformed data.

POLYNOMIAL REGRESSION



POLYNOMIAL REGRESSION

Why Polynomial Regression is Still Linear Regression

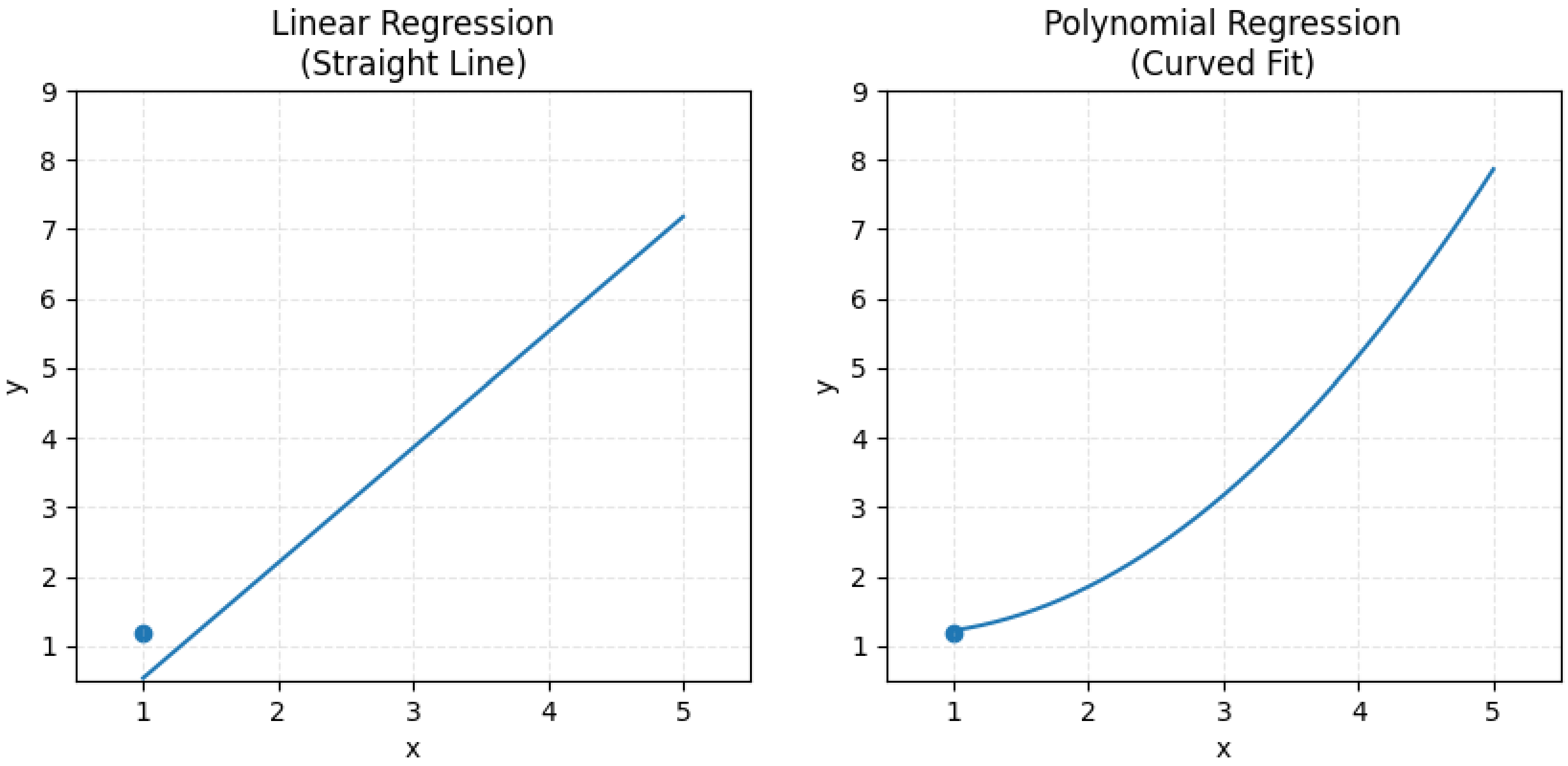
- Because the model is **linear in parameters (weights)**.

$$\hat{y} = w_0 + w_1 x + w_2 x^2$$

- Linear in w_0, w_1, w_2
  Non-linear only in variable x
- **Polynomial regression is linear in parameters but non-linear in features.**

POLYNOMIAL REGRESSION

Why Polynomial Regression is Still Linear Regression

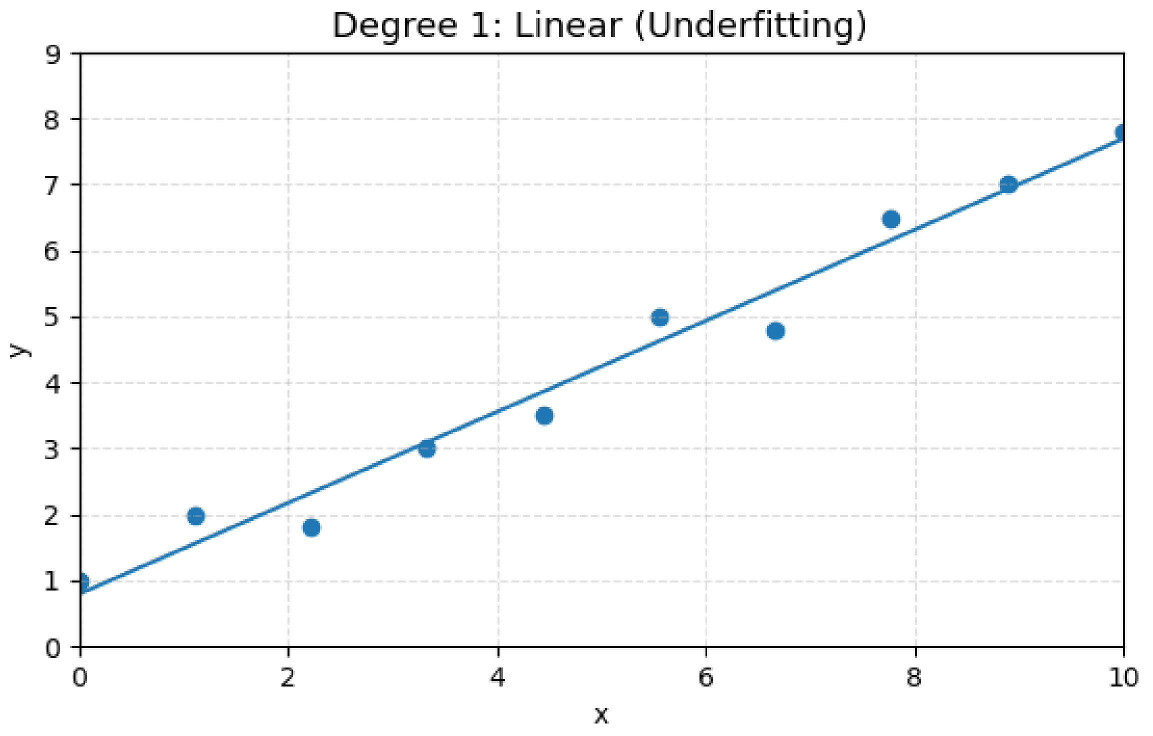


Polynomial regression allows the model to **bend** to fit data.

POLYNOMIAL REGRESSION

Degree of Polynomial and Its Effect

Degree	Shape	Effect
1	Line	Underfitting
2	Parabola	Good fit
3	Cubic	More flexibility
High	Highly wiggly	Overfitting



POLYNOMIAL REGRESSION

EXAMPLE

Problem

- A company studies the relationship between **experience (x)** and **salary (y)**.

x (Years)	y (₹ Lakhs)
1	3
2	5
3	9
4	15

The trend is **non-linear**.
We choose a **2nd-degree polynomial model**.

POLYNOMIAL REGRESSION

EXAMPLE

Problem

- A company studies the relationship between **experience (x)** and **salary (y)**.
- **Step 1: Assume the Model**

$$\hat{y} = w_0 + w_1 x + w_2 x^2$$

- **Step 2: Transform Input Features**

x	x^2
1	1
2	4
3	9
4	16

$x \rightarrow x^2$

Polynomial feature

POLYNOMIAL REGRESSION

EXAMPLE

Problem

Step 3.2: Add Bias Term (Column of 1s)

- Bias term corresponds to w_0.
- $X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix}$
- Each row represents:

$$[1,\ x,\ x^2]$$

Step 3.3: Output Vector

- $Y = \begin{bmatrix} 3 \\ 5 \\ 9 \\ 15 \end{bmatrix}$

STEP 4: WRITE THE NORMAL EQUATION

The goal of regression is to **minimize Mean Squared Error**.

The closed-form solution is:

$$\boxed{W = (X^T X)^{-1} X^T Y}$$

Where:

$$W = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix}$$

POLYNOMIAL REGRESSION

EXAMPLE

Problem

STEP 5: CALCULATE $X^T X$

Step 5.1: Write X^T

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 4 & 9 & 16 \end{bmatrix}$$

Step 5.2: Multiply $X^T \times X$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 4 & 9 & 16 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix}$$

Final $X^T X$

$$X^T X = \begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix}$$

Step 5.3: Compute Each Element

First Row

- (1,1): $1 + 1 + 1 + 1 = 4$
- (1,2): $1 + 2 + 3 + 4 = 10$
- (1,3): $1 + 4 + 9 + 16 = 30$

Second Row

- (2,1): $1 + 2 + 3 + 4 = 10$
- (2,2): $1 + 4 + 9 + 16 = 30$
- (2,3): $1 + 8 + 27 + 64 = 100$

Third Row

- (3,1): $1 + 4 + 9 + 16 = 30$
- (3,2): $1 + 8 + 27 + 64 = 100$
- (3,3): $1 + 16 + 81 + 256 = 354$

POLYNOMIAL REGRESSION

EXAMPLE

Problem

STEP 6: CALCULATE X^TY

$$X^TY = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 4 & 9 & 16 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 9 \\ 15 \end{bmatrix}$$

Step 6.1: Compute Each Element

First element:

$$3 + 5 + 9 + 15 = 32$$

Second element:

$$(1)(3) + (2)(5) + (3)(9) + (4)(15) = 3 + 10 + 27 + 60 = 100$$

Third element:

$$(1)(3) + (4)(5) + (9)(9) + (16)(15) = 3 + 20 + 81 + 240 = 344$$

Final X^TY

$$X^TY = \begin{bmatrix} 32 \\ 100 \\ 344 \end{bmatrix}$$

POLYNOMIAL REGRESSION

EXAMPLE

Problem

STEP 7: COMPUTE $\left((X^T X)^{-1}\right)$

Given Matrix

$$X^T X = \begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix}$$

Important Formula (Memorize)
For any square matrix A:

$$A^{-1} = \frac{1}{|A|} \cdot adj(A)$$

Where:
$|A|$= **Determinant**
adj(A) =**Adjoint matrix**

POLYNOMIAL REGRESSION

EXAMPLE

Problem

STEP 7.1: FIND THE DETERMINANT $|X^T X|$

For a 3×3 matrix:

$$|A| = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Substitute values:

$$|X^T X| = \begin{vmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{vmatrix}$$

$$= 4(30 \cdot 354 - 100 \cdot 100)$$
$$- 10(10 \cdot 354 - 100 \cdot 30)$$
$$+ 30(10 \cdot 100 - 30 \cdot 30)$$

Compute each term carefully:

First term:

$$4(10620 - 10000) = 4(620) = 2480$$

Second term:

$$-10(3540 - 3000) = -10(540) = -5400$$

Third term:

$$30(1000 - 900) = 30(100) = 3000$$

Final Determinant

$$|X^T X| = 2480 - 5400 + 3000 = 80$$

Determinant ≠ 0, so inverse **exists**

POLYNOMIAL REGRESSION

EXAMPLE

Problem

STEP 7.2: FIND THE COFACTOR MATRIX

We compute **minors and cofactors** element by element.

Cofactor C_{11}

Remove row 1, column 1:

$$\begin{vmatrix} 30 & 100 \\ 100 & 354 \end{vmatrix} = (30)(354) - (100)(100) = 620$$

$$C_{11} = +620$$

Cofactor C_{12}

$$\begin{vmatrix} 10 & 100 \\ 30 & 354 \end{vmatrix} = 3540 - 3000 = 540$$

$$C_{12} = -540$$

Cofactor C_{13}

$$\begin{vmatrix} 10 & 30 \\ 30 & 100 \end{vmatrix} = 1000 - 900 = 100$$

$$C_{13} = +100$$

Cofactor C_{21}

$$\begin{vmatrix} 10 & 30 \\ 100 & 354 \end{vmatrix} = 3540 - 3000 = 540$$

$$C_{21} = -540$$

Cofactor C_{22}

$$\begin{vmatrix} 4 & 30 \\ 30 & 354 \end{vmatrix} = 1416 - 900 = 516$$

$$C_{22} = +516$$

Cofactor C_{23}

$$\begin{vmatrix} 4 & 10 \\ 30 & 100 \end{vmatrix} = 400 - 300 = 100$$

$$C_{23} = -100$$

POLYNOMIAL REGRESSION

EXAMPLE

Problem

STEP 7.2: FIND THE COFACTOR MATRIX

We compute **minors and cofactors** element by element.

Cofactor C_{31}

$$\begin{vmatrix} 10 & 30 \\ 30 & 100 \end{vmatrix} = 1000 - 900 = 100$$

$$C_{31} = +100$$

Cofactor C_{32}

$$\begin{vmatrix} 4 & 30 \\ 10 & 100 \end{vmatrix} = 400 - 300 = 100$$

$$C_{32} = -100$$

Cofactor C_{33}

$$\begin{vmatrix} 4 & 10 \\ 10 & 30 \end{vmatrix} = 120 - 100 = 20$$

$$C_{33} = +20$$

POLYNOMIAL REGRESSION

EXAMPLE

Problem

STEP 7.3: FORM THE COFACTOR MATRIX

$$C = \begin{bmatrix} 620 & -540 & 100 \\ -540 & 516 & -100 \\ 100 & -100 & 20 \end{bmatrix}$$

STEP 7.4: COMPUTE THE ADJOINT

Adjoint = **Transpose of Cofactor Matrix**

$$\text{adj}(X^T X) = \begin{bmatrix} 620 & -540 & 100 \\ -540 & 516 & -100 \\ 100 & -100 & 20 \end{bmatrix}$$

(Matrix is symmetric, so transpose is same)

STEP 7.5: COMPUTE THE INVERSE

$$(X^T X)^{-1} = \frac{1}{|X^T X|} \cdot adj(X^T X)$$

$$= \frac{1}{80} \begin{bmatrix} 620 & -540 & 100 \\ -540 & 516 & -100 \\ 100 & -100 & 20 \end{bmatrix}$$

Divide Each Element by 80

$$(X^T X)^{-1} = \begin{bmatrix} 7.75 & -6.75 & 1.25 \\ -6.75 & 6.45 & -1.25 \\ 1.25 & -1.25 & 0.25 \end{bmatrix}$$

POLYNOMIAL REGRESSION

EXAMPLE

Problem

STEP 8: COMPUTE THE REGRESSION COEFFICIENTS (FINAL OUTPUT)
Formula Used (Normal Equation)

$$\boxed{\mathbf{W} = \left(X^T X\right)^{-1} X^T \mathbf{y}}$$

Where

$$\mathbf{W} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix}$$

POLYNOMIAL REGRESSION

EXAMPLE

Problem

Step 8.1: Known Values from Previous Steps
Exact inverse (already computed)

$$(X^T X)^{-1} = \begin{bmatrix} 7.75 & -6.75 & 1.25 \\ -6.75 & 6.45 & -1.25 \\ 1.25 & -1.25 & 0.25 \end{bmatrix}$$

Exact $X^T y$vector (from Step 6)

$$X^T y = \begin{bmatrix} 20 \\ 70 \\ 244 \end{bmatrix}$$

This comes from column-wise multiplication of X^T with output vector y; assumed already derived earlier in your solution.

Step 8.2: Matrix Multiplication

$$\mathbf{W} = \begin{bmatrix} 7.75 & -6.75 & 1.25 \\ -6.75 & 6.45 & -1.25 \\ 1.25 & -1.25 & 0.25 \end{bmatrix} \begin{bmatrix} 20 \\ 70 \\ 244 \end{bmatrix}$$

POLYNOMIAL REGRESSION

EXAMPLE

Problem

Step 8.3: Compute Each Weight Explicitly

Bias term w_0

$$\begin{aligned} w_0 &= (7.75)(20) + (-6.75)(70) + (1.25)(244) \\ &= 155 - 472.5 + 305 \\ &= \boxed{-12.5} \end{aligned}$$

Coefficient w_1

$$\begin{aligned} w_1 &= (-6.75)(20) + (6.45)(70) + (-1.25)(244) \\ &= -135 + 451.5 - 305 \\ &= \boxed{11.5} \end{aligned}$$

Coefficient w_2

$$\begin{aligned} w_2 &= (1.25)(20) + (-1.25)(70) + (0.25)(244) \\ &= 25 - 87.5 + 61 \\ &= \boxed{-1.5} \end{aligned}$$

POLYNOMIAL REGRESSION

EXAMPLE

Problem

FINAL EXACT OUTPUT

$$\boxed{\mathbf{W} = \begin{bmatrix} -12.5 \\ 11.5 \\ -1.5 \end{bmatrix}}$$

Final Multiple Linear Regression Equation

$$\boxed{\hat{y} = -12.5 + 11.5x_1 - 1.5x_2}$$

Where:
x_1 =First input feature
x_2 =Second input feature
$\hat{y}$ =Predicted output

Term	Meaning
(w_0)	Base value
(w_1)	Initial trend
(w_2)	Acceleration

POLYNOMIAL REGRESSION

EXAMPLE

Problem

$$\boxed{\hat{y} = -12.5 + 11.5x_1 - 1.5x_2}$$

Interpretation of the Bias Term ($w_0 = -12.5$)
Mathematical Meaning

$$w_0 = -12.5$$

This is the **intercept** of the regression plane.

When $x_1 = 0$and $x_2 = 0$:

$$\hat{y} = -12.5$$

It is the **starting value** of the prediction.
It shifts the regression plane **up or down**.
Negative bias means the plane starts **below the origin**.

POLYNOMIAL REGRESSION

EXAMPLE

$$\boxed{\hat{y} = -12.5 + 11.5x_1 - 1.5x_2}$$

Problem

Interpretation of Coefficient $w_1 = 11.5$
Mathematical Meaning

$$w_1 = 11.5$$

Keeping x_2 constant,
If x_1 increases by 1 unit, then $\hat{y}$ **increases by 11.5 units**.

Intuition
x_1 has a **strong positive influence** on the output.

The regression plane **tilts steeply upward** along x_1.

Increase x1 ⟶ y increases sharply
For a unit increase in x_1 ,the predicted output increases by 11.5 units, assuming other variables remain constant.

POLYNOMIAL REGRESSION

EXAMPLE

Problem

$$\boxed{\hat{y} = -12.5 + 11.5x_1 - 1.5x_2}$$

Interpretation of Coefficient $w_2 = -1.5$
Mathematical Meaning

$$w_2 = -1.5$$

Keeping x_1 constant,
If x_2 **increases by 1 unit**, then $\hat{y}$ **decreases by 1.5 units**.
Intuition
x_2 has a **negative effect** on the output.
Increasing x_2 slightly **pulls the plane downward**.

Increase x2 ⟶ y decreases slightly

The negative coefficient indicates an inverse relationship between x_2 and the output variable.

POLYNOMIAL REGRESSION

EXAMPLE

Problem

$$\boxed{\hat{y} = -12.5 + 11.5x_1 - 1.5x_2}$$

Relative Importance of Features

Compare magnitudes:

Feature	Coefficient	Effect
(x_1)	+11.5	Strong positive influence
(x_2)	−1.5	Weak negative influence

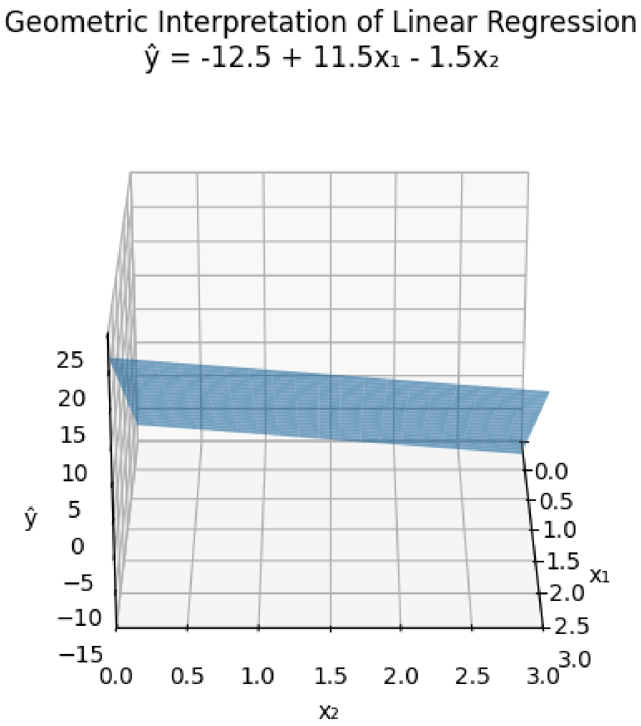
x_1 is the **dominant predictor**.

x_2 has a **smaller, opposing effect**.

The magnitude of coefficients indicates that x_1 contributes more significantly to the prediction than x_2.

POLYNOMIAL REGRESSION

Model-Level Interpretation



Suppose:
- $x_1 = 4$
- $x_2 = 3$

$$\hat{y} = -12.5 + (11.5)(4) - (1.5)(3)$$
$$\hat{y} = -12.5 + 46 - 4.5 = \boxed{29}$$

Meaning
- A strong increase from x_1
- Slight reduction due to x_2
- Final predicted output = **29**

- The system **depends mainly on** x_1.
- $x_2$acts as a **penalty or controlling factor**.
- The relationship is **linear and additive**.

The multiple linear regression model indicates that the output increases significantly with x_1 , decreases slightly with x_2 ,and has a negative baseline value when all predictors are zero.

POLYNOMIAL REGRESSION

Advantages of Polynomial Regression

- Models curved relationships
  Simple extension of linear regression
  Interpretable for low degree

Limitations

- Overfitting if degree is too high
  Sensitive to outliers
  Poor extrapolation beyond data

Real-World Applications

- Learning curves
- Economic growth
- Sensor calibration
- Medical dosage response

Bias-Variance Trade off

The **Bias-Variance Tradeoff** is one of the **most fundamental concepts in machine learning**, helping us understand **why models make errors** and how to **improve predictions**.

Sources of Error in Machine Learning

- When we train a machine learning model, the **prediction error** for a new unseen point can be decomposed into three components:

$$\text{Total } Error = Bias^2 + Variance + Irreducible\ Error$$

Where:

Term	Meaning	Example
Bias²	Error due to **wrong assumptions** in the model	Trying to fit a straight line to curved data
Variance	Error due to **sensitivity to small changes** in training data	Model fits perfectly to one dataset but fails on another
Irreducible Error (ε)	Noise inherent in the data	Random fluctuations, measurement errors

Bias-Variance Trade off

Bias

- **Bias** measures how much the **average prediction** of the model differs from the **true underlying function**.

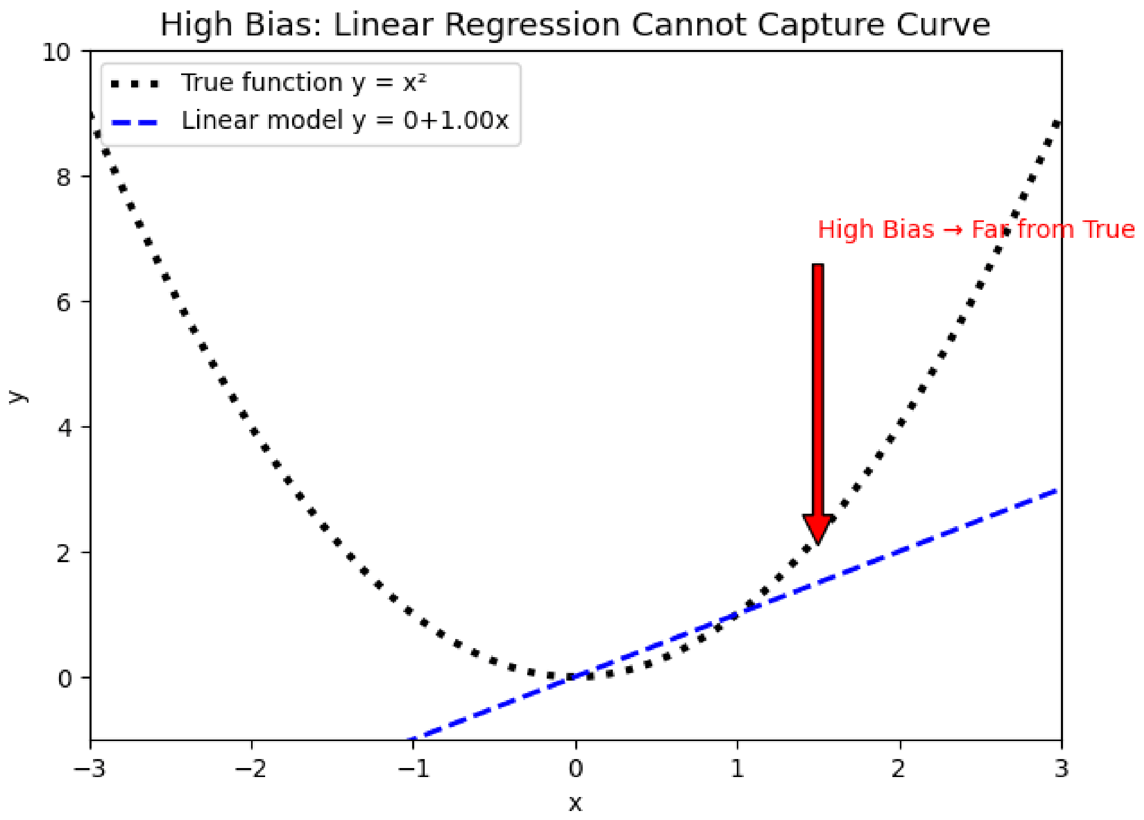
  Bias = ***Error from assumptions made by the model***

- High bias → Model is **too simple**, **underfits** data
- Low bias → Model can **capture complex patterns**
- Example:

True function: $y = x^2$

Model: Linear Regression $y = w_0 + w_1 x$

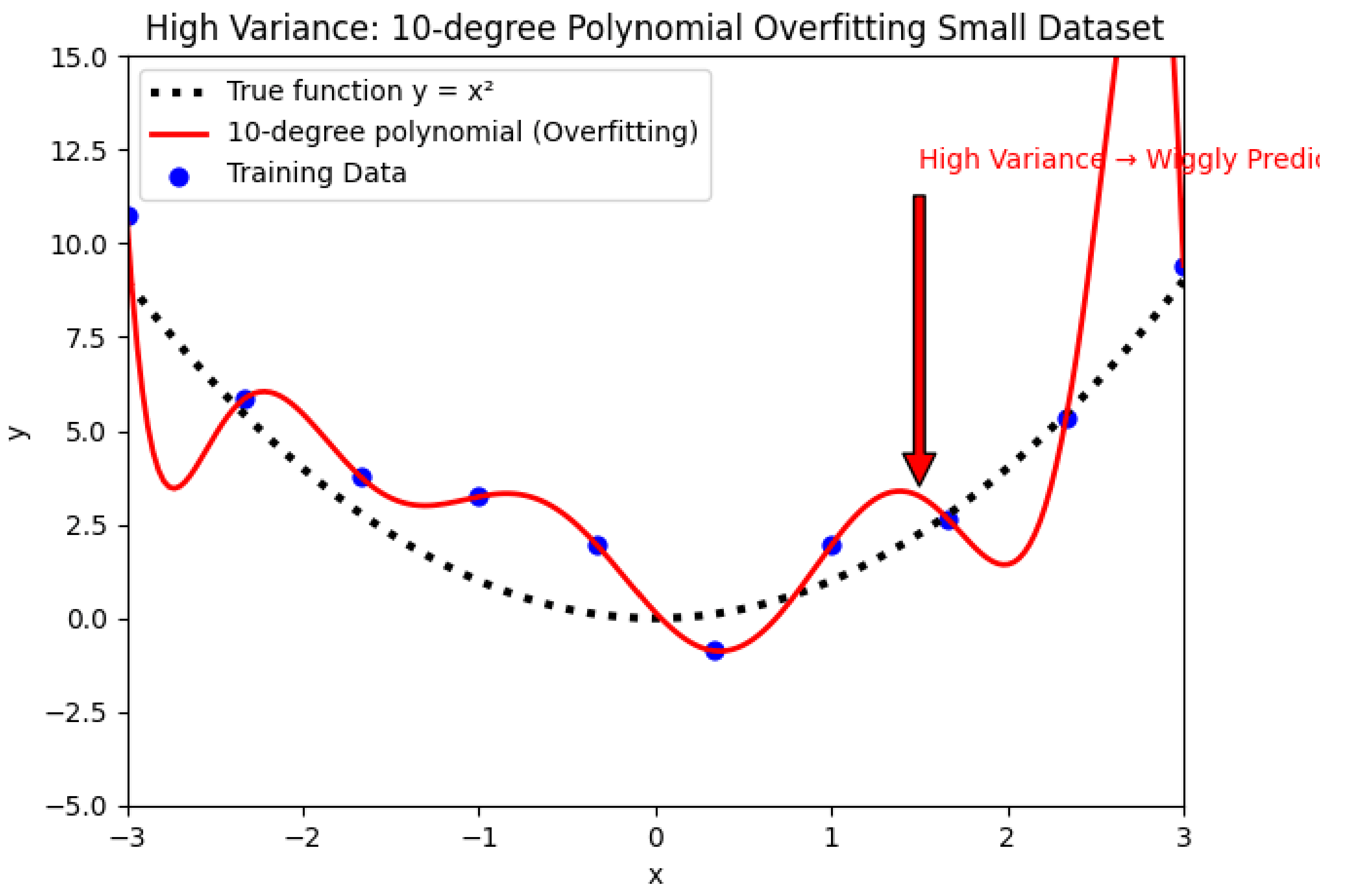
Linear model cannot capture the curve → **high bias**



Bias-Variance Trade off

Variance

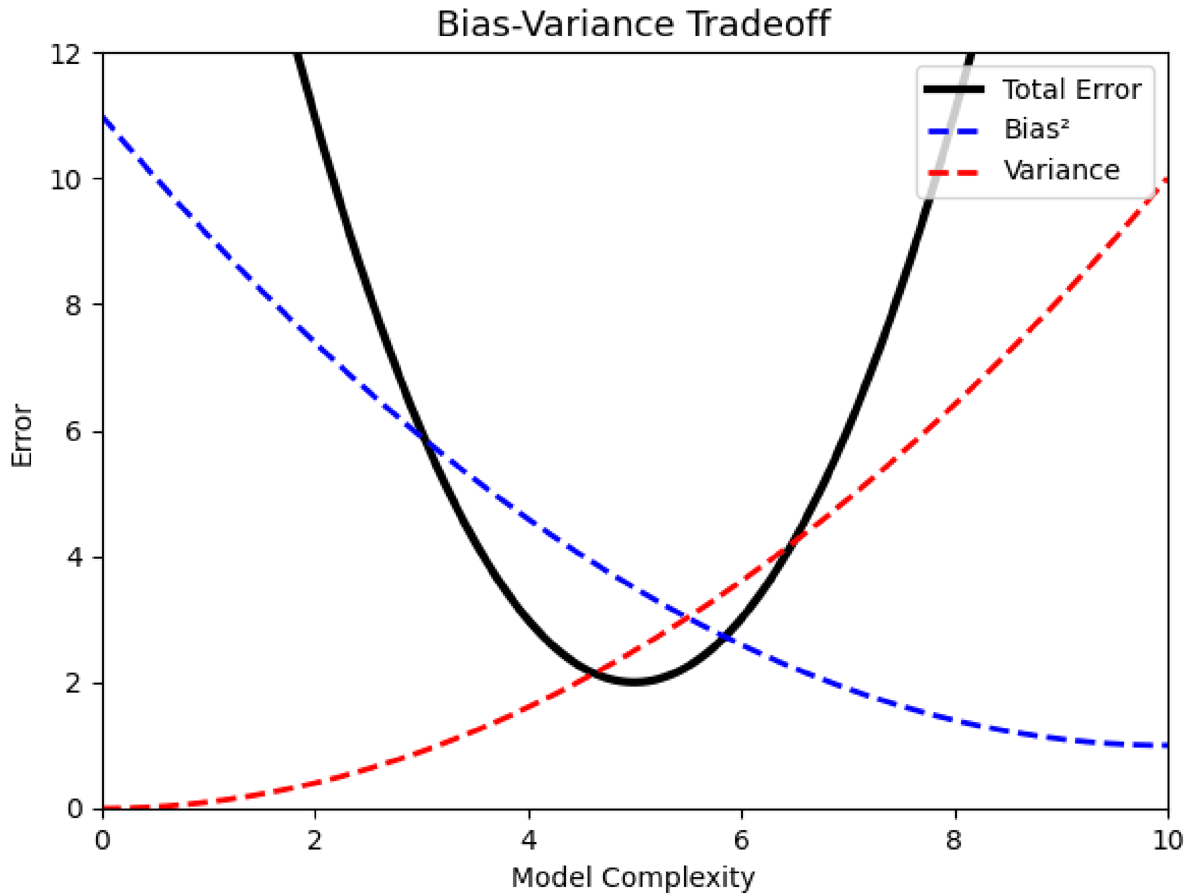
- **Variance** measures how much the **model prediction changes** if we train it on **different datasets** from the same population.
- High variance → Model is **too sensitive**, **overfits** the data
- Low variance → Model predictions are **stable**



Bias-Variance Trade off

Bias-Variance Tradeoff

- You cannot minimize **bias and variance simultaneously**. Reducing one often **increases the other**.



- **Low complexity** → High bias, low variance
- **High complexity** → Low bias, high variance
- **Optimal complexity** → Minimum total error

Bias-Variance Trade off

Example Scenario

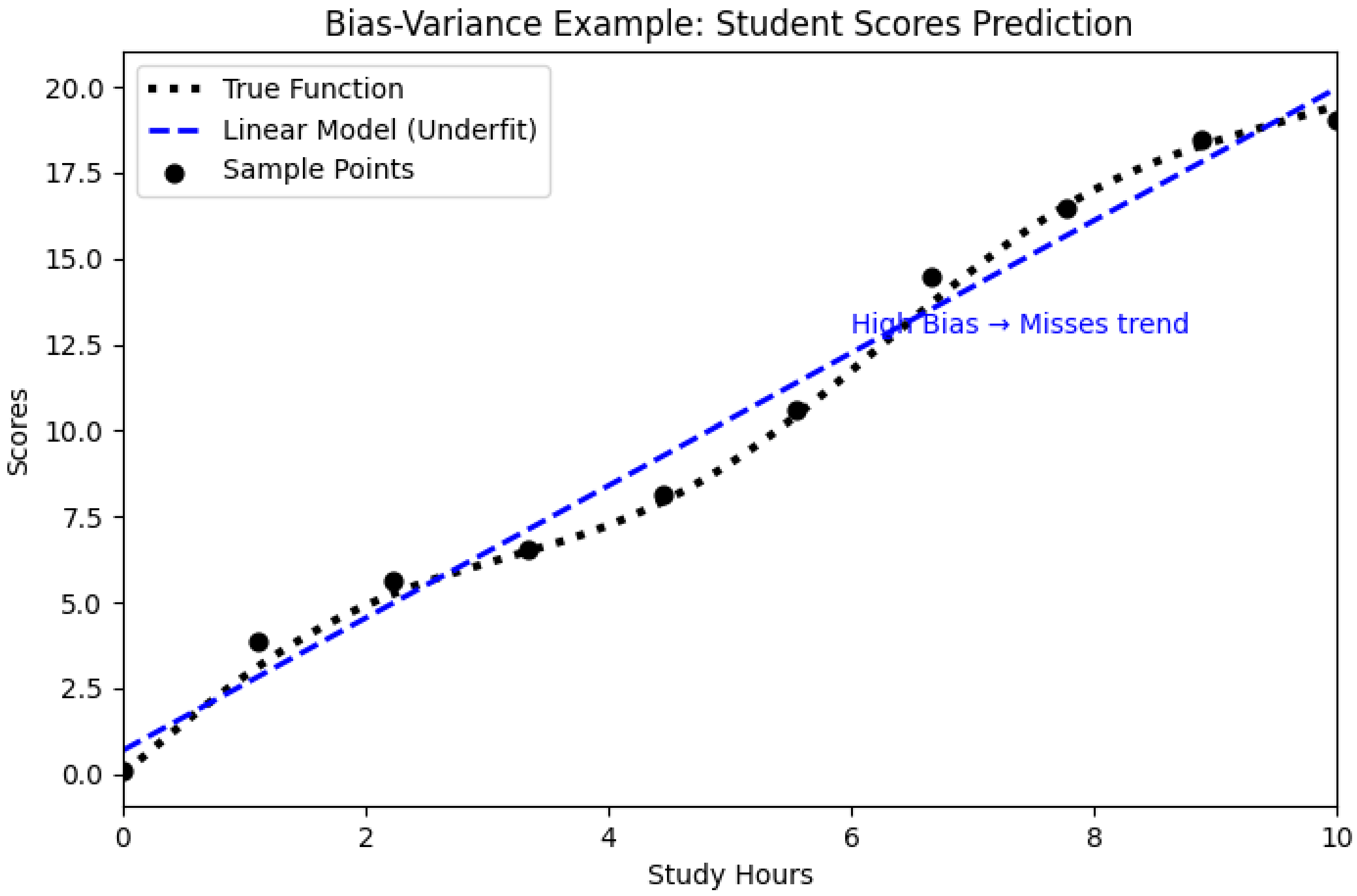
- Suppose we want to predict **student scores based on study hours**.

Model Type	Complexity	Bias	Variance	Under/Overfit?
Linear Regression	Low	High	Low	Underfit
Quadratic Regression	Medium	Medium	Medium	Balanced
10th Degree Polynomial	Very High	Low	Very High	Overfit

Bias-Variance Trade off

Example Scenario

- Suppose we want to predict **student scores based on study hours**.



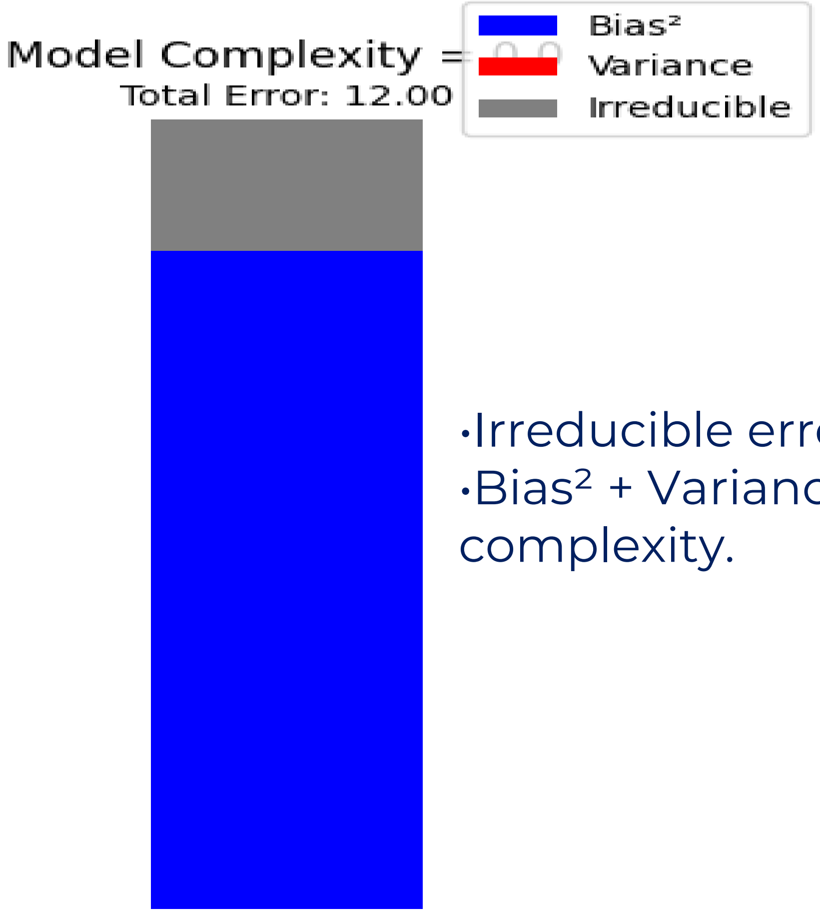
- Linear: Misses curve → high bias
- Quadratic: Captures trend → balanced
- High polynomial: Fits every point → high variance

Bias-Variance Trade off

How Bias and Variance Affect Error

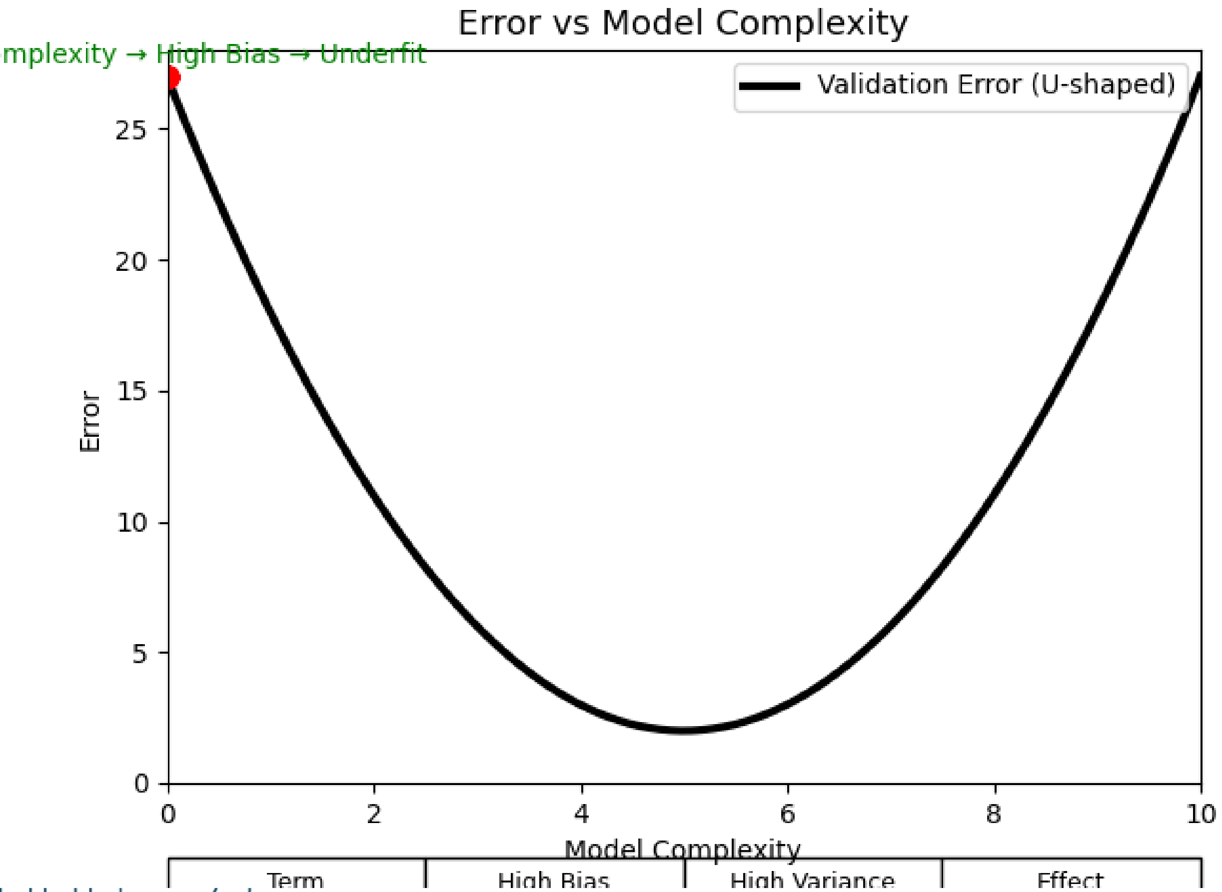
$$\text{Expected } Error = \underbrace{Bias^2}_{\textit{error from wrong model}} + \underbrace{Variance}_{\textit{error from data fluctuations}} + \underbrace{\sigma^2}_{\textit{irreducible noise}}$$

Bias-Variance Trade off



- Irreducible error = cannot be improved
- Bias² + Variance = can be reduced by choosing right model complexity.

Bias-Variance Trade off



Term	High Bias	High Variance	Effect
Prediction	Too simple	Too wiggly	Total error high
Training	Bad fit	Perfect fit	Underfit / Overfit
Solution	Increase complexity	Reduce complexity / regularize	Balanced model

Bias-Variance Trade off

1. **Simple models** → High bias, low variance → Underfit
2. **Complex models** → Low bias, high variance → Overfit
3. **Goal** → Balance complexity for **minimum total error**
4. **Regularization** (like Ridge, Lasso) can reduce variance
5. **Cross-validation** helps find optimal complexity

- Plot **training vs validation error**:
  - **Training error low, validation high** → Overfit → reduce complexity
  - **Both errors high** → Underfit → increase complexity
- Use **polynomial degree**, **number of features**, and **regularization** to balance bias-variance